

Food Analysis and Animal Feed Biotechnology

Science

May, 2026

Food Analysis and Animal Feed Biotechnology (A36344)

Short Title:	Food Analysis Animal Feed Bio	
Faculty	Science and Computing	
Department	Science	
Cluster	Food Science	
Author	JFRISBY	
Credits	5	Level: Intermediate

Description of Module / Aims

This module introduces the student to basic food analysis. Novel food packaging and the biological aspects of animal feed are introduced.

Programmes

AGRI-0103 BSc in Agriculture (WD_SAGUL_D)

3 / 5 / M

Indicative Content

- Laboratory safety and safe disposal of chemicals
- Food analysis tests, including some microbiological analysis
- An introduction to probiotics, prebiotics and zenobiotics in a range of animal feeds (including animal clinical trials)
- Innovative application of microorganisms as novel food packaging systems
- The marine environment and the agri/food industry
- GMO`s in the agri/food sector

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use safe laboratory procedures including the correct disposal of laboratory waste.
2. Complete proximate and microbiological analysis on a range of food products and animal feeds.
3. Communicate results from report laboratory experiments in an appropriate format and evaluate the data.
4. Analyse various Food Analysis experiments and their uses in the food industry.
5. Differentiate the various types of animal feeds and their microbial/non-microbial composition.
6. Analyse regulations, food safety concerns of GMO`s and the basics of animal clinical trials.

Learning and Teaching Methods

- Practical laboratory work.
- Group work.
- Lectures and discussion.
- Blended learning.
- Case studies.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Practical</i>	50%	1,2,3
<i>In-Class Assessment</i>	50%	4,5,6

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- "Food Safety Authority of Ireland." www.fsai.ie
- "Module, Food Analysis and Animal Feed Biotechnology." www.moodle.wit.ie

Requested Resources

- Room Type: Biology Lab